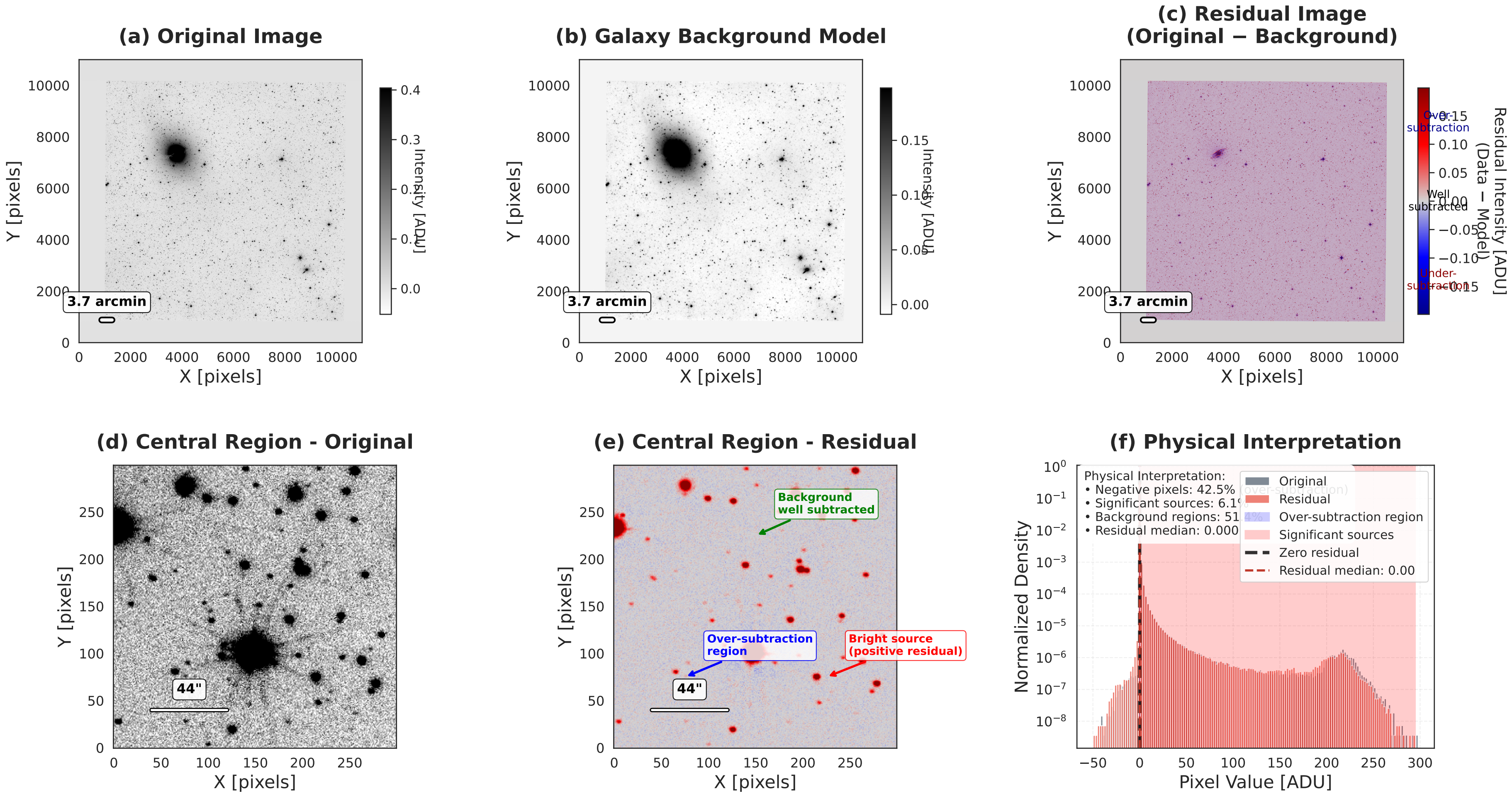


Galaxy Background Subtraction Physics: CenA11 F660 (H α)
Unsharp Mask: Residual = Data – Gaussian(Median(Data))



Physical Interpretation of Residual Values

PHYSICAL INTERPRETATION OF RESIDUAL VALUES

The residual image shows: Data – Galaxy Background Model

- ☐ NEGATIVE VALUES (Over-subtraction):
 - 42.5% of pixels are negative
 - Occur where background model > actual data
 - Common around bright stars and sharp gradients
 - Expected in unsharp mask processing
- VALUES NEAR ZERO (Well subtracted):
 - 98.4% of pixels near zero (± 0.1 ADU)
 - Background successfully removed
 - Represents clean noise floor
- ☐ POSITIVE VALUES (Under-subtraction/Real sources):
 - 42.5% of pixels are positive
 - Real astronomical sources (stars, clusters)
 - Regions where model < actual background
 - Used for robust photometry measurements
- ☐ Statistical Summary:
 - Median residual: 0.0000 ADU
 - Standard deviation: 0.0196 ADU
 - Dynamic range: -49.72 to 294.92 ADU